

Internet Performance Analytics System

SQL Server & Power BI Portfolio Project

**Prepared by
Efe Elmas**

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Description of the Project

This project focuses on building a relational database and reporting system for analyzing internet performance data. The main purpose of the project is to store information about users, their locations, devices, internet sessions, routers, network performance logs, and connection errors in a structured database system.

The database was implemented in MSSQL Server. It consists of multiple related tables connected through primary key and foreign key relationships. After creating the tables and inserting the sample data, I tested the relationships between the tables by using SQL JOIN queries. I also created SQL views and stored procedures to make some of the repeated analysis queries easier to access.

The reporting part of the project was prepared in Power BI. I created seven report pages to analyze the dataset from different perspectives. These pages focus on general network performance, location-based analysis, subscription type comparison, device analysis, time-based performance changes, error analysis, and reliability summary. Through these reports, the project aims to show how relational data stored in SQL Server can be used for visual analysis in Power BI.

About the Dataset

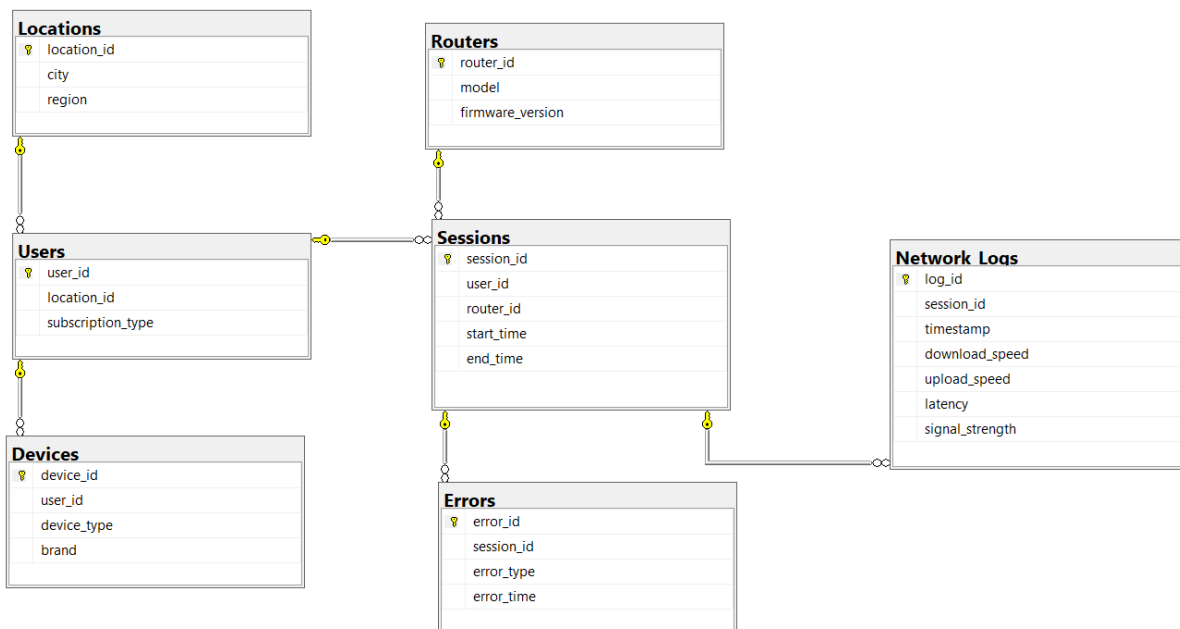
The dataset was created as a synthetic relational dataset for an internet performance analysis scenario. Since real-world internet service provider or telecom performance data is not publicly accessible for this project, I generated sample data based on realistic assumptions.

The main purpose of the dataset is to create a relational structure that allows SQL JOIN operations and Power BI reports. Therefore, the tables were designed to be connected through primary key and foreign key relationships. The central table of the database is the **Sessions Table**, since it connects users, routers, network performance logs, and error records.

Schema of the Dataset

The dataset consists of seven tables: Locations, Users, Devices, Routers, Sessions, Network_Logs, and Errors. The Locations table stores city and region information. The Users table is connected to Locations through location_id. The Devices table is connected to Users through user_id. The Sessions table connects users and routers by using user_id and router_id. Network_Logs and Errors are both connected to Sessions through session_id.

The ERD diagram of the dataset:



The main relationships in the database are:

- Locations → Users
- Users → Devices
- Users → Sessions
- Routers → Sessions
- Sessions → Network_Logs
- Sessions → Errors

This schema makes it possible to analyze internet performance data from different perspectives such as location, subscription type, device information, router model, time, and connection errors.

Datatypes of the Dataset

The main datatypes used in the dataset are INT, NVARCHAR, DATETIME, and FLOAT.

INT is mainly used for ID columns such as user_id, session_id, router_id, log_id, and error_id. These columns are used as primary keys and foreign keys, and most of them are defined with IDENTITY(1,1).

NVARCHAR(50) is used for text-based columns such as city, region, subscription_type, device_type, brand, router model, firmware version, and error type.

DATETIME is used for time-related columns such as start_time, end_time, timestamp, and error_time.

FLOAT is used for performance metrics such as download_speed, upload_speed, and latency. In addition, signal_strength is stored as INT because it is represented with whole number values.

I also used several constraints to improve data consistency, such as CHECK constraints for download speed and latency, and a DEFAULT constraint for subscription_type.

Description of the Dataset

The dataset consists of seven relational tables. Each table represents a different part of the internet performance system. The row counts of the tables are shown below.

Table	Description	Number of Rows
Locations	Stores city and region information	10
Users	Stores user subscription and location information	50
Devices	Stores device type and brand information for users	53
Routers	Stores router model and firmware version information	12
Sessions	Stores user internet connection sessions	100
Network_Logs	Stores network performance measurements	277
Errors	Stores connection error records	43

The Locations table contains the cities and regions used in the project.

The Users table stores user information. Each user has a location_id and a subscription_type. The subscription types are Basic and Premium. This table is useful for analyzing performance differences between user groups.

The Devices table stores device-related information. It includes device type and brand information. Since users may have more than one device, this table is connected to the Users table.

The Routers table stores router model and firmware version information. Router data is used to analyze whether router models are related to connection quality or errors.

The Sessions table stores internet connection sessions. Each session is connected to a user and a router. This table is one of the most important tables in the database because it connects users, routers, network logs, and errors.

The Network_Logs table stores performance measurements for internet sessions. It includes timestamp, download_speed, upload_speed, latency, and signal_strength columns. This table is the main table for performance analysis.

The Errors table stores connection problems related to sessions. It includes different error types such as Weak Signal, High Latency, Slow Connection, Timeout, and Disconnect. This table is used for error analysis and reliability reporting.

Views and Stored Procedures

In this project, views and stored procedures are mostly used to make SQL queries more organized and reusable. The views mainly combine columns from different related tables by using JOIN statements. Some of them also include GROUP BY statements to summarize the data by city, region, or subscription type.

Stored procedures are used to retrieve specific results more easily. Some procedures return general information, while others use parameters such as user ID, city, or error type.

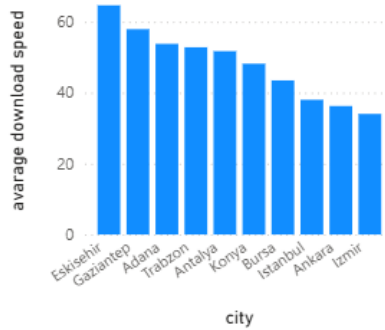
I did not directly use the views and stored procedures in Power BI. However, they were useful for testing the relationships between tables and seeing the joined results clearly in SQL Server.

Description of the Reports

Report Page 1. Network Performance Overview

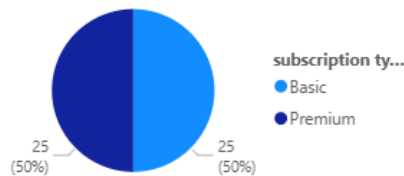
In this report page, I wanted to give a general overview of the internet performance dataset. The page includes different visuals that summarize network performance, user subscription distribution, city-based performance, and connection errors.

Average Download Speed by City

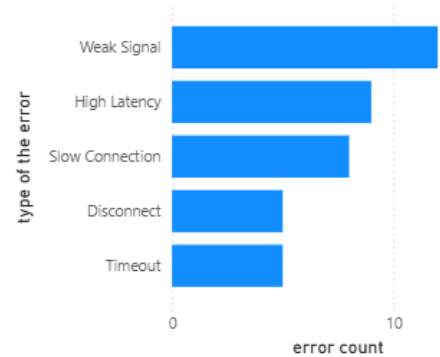


NETWORK PERFORMANCE OVERVIEW

Users by Subscription Type



Errors by Type



City Performance Summary

city	average download speed	number of errors	average latency
Izmir	34,03	8	66,74
Ankara	36,26	8	63,19
Bursa	43,48	6	56,24
Istanbul	38,02	8	52,71
Antalya	51,73	4	42,33
Trabzon	52,83	1	38,22
Konya	48,20	1	38,07
Gaziantep	57,93	2	29,67
Adana	53,80		28,53
Eskisehir	64,67	1	25,33
Sum	45,13	39	48,69

Error Count Over Time



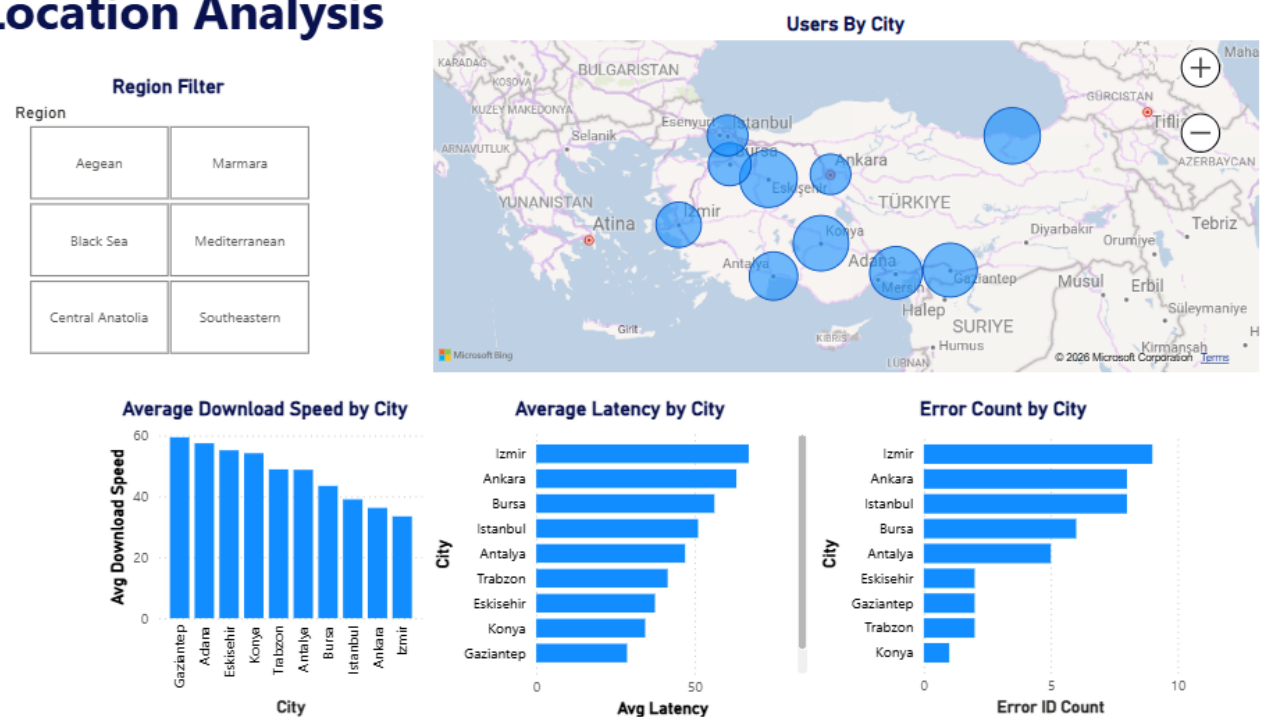
For city-based performance, I used the Locations and Network_Logs tables to show the average download speed by city. I also used the Errors table to show the distribution of error types and the change in error count over time. The Users table was used to create the subscription type chart, which shows the distribution of Basic and Premium users.

The report page includes a column chart, bar chart, donut chart, line chart, and a summary table. The summary table shows city-based average download speed, average latency, and error count. Overall, this page works as the main overview page of the project before moving into more detailed report pages.

Report Page 2. Location Analysis

In this report page, I focused on location-based network analysis. The main purpose of this page is to compare internet performance and connection errors across different cities and regions.

Location Analysis

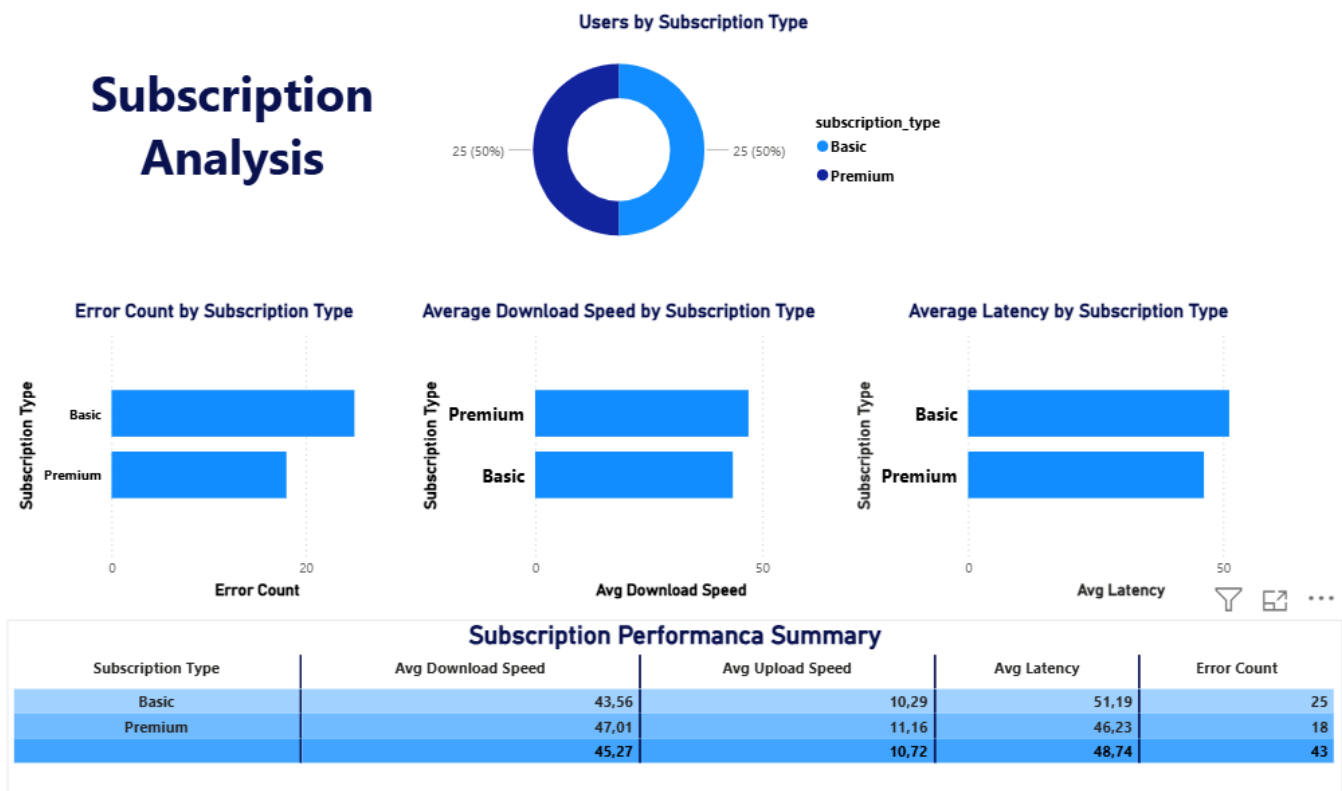


I used the Locations table together with Users, Sessions, Network_Logs, and Errors tables. The map visual shows the distribution of users by city. In addition, I used city information to compare average download speed, average latency, and error count by city.

The report page includes a map, column charts, bar charts, and a region slicer. The region slicer allows the report to be filtered by geographical region. This page helps to understand whether internet performance differs depending on location

Report Page 3. Subscription Analysis

In this report page, I analyzed the differences between Basic and Premium users. The main purpose of this page is to compare subscription types in terms of network performance and error count.



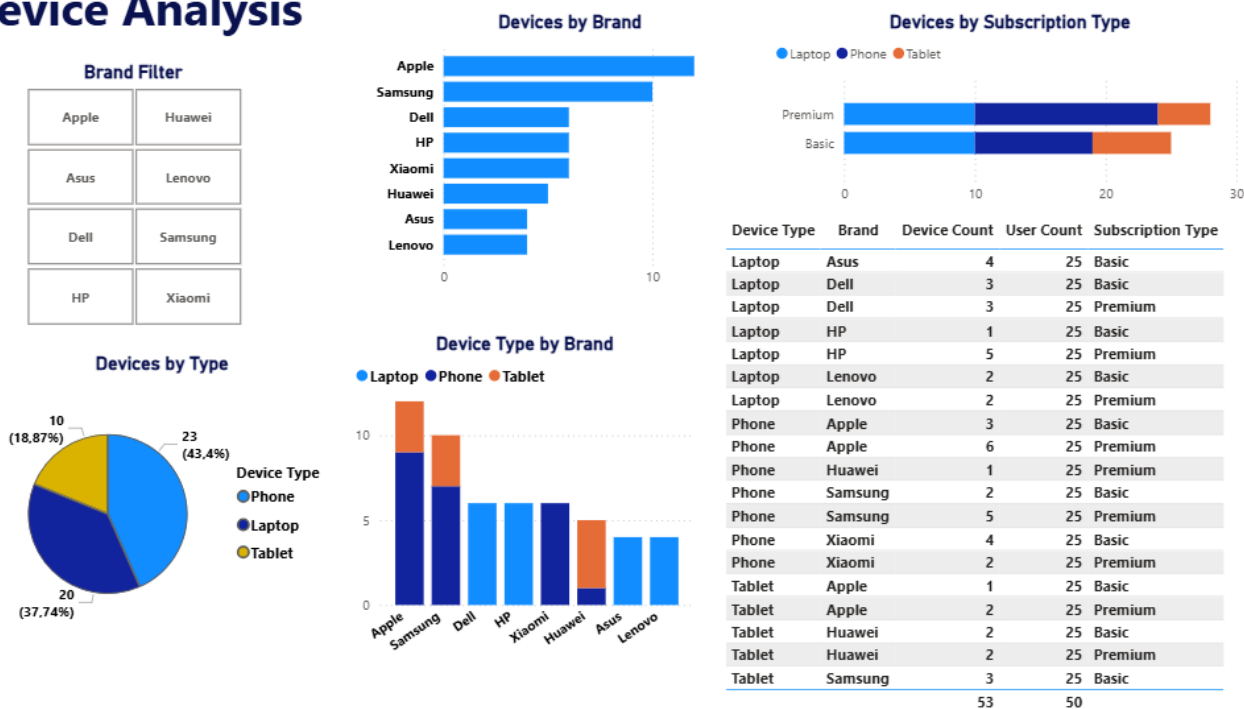
I mainly used the Users, Sessions, Network_Logs, and Errors tables. The subscription type column from the Users table was used as the main category. I compared average download speed, average latency, and error count according to subscription type. I also added a donut chart to show the distribution of Basic and Premium users.

The report page includes bar charts, a donut chart, and a summary table. The summary table shows average download speed, average upload speed, average latency, and error count for each subscription type. According to this page, subscription type can be analyzed as one of the main dimensions affecting user experience.

Report Page 4. Device Analysis

In this report page, I focused on device-related information in the dataset. The main purpose of this page is to analyze the distribution of devices by type and brand, and to see how device information relates to users.

Device Analysis



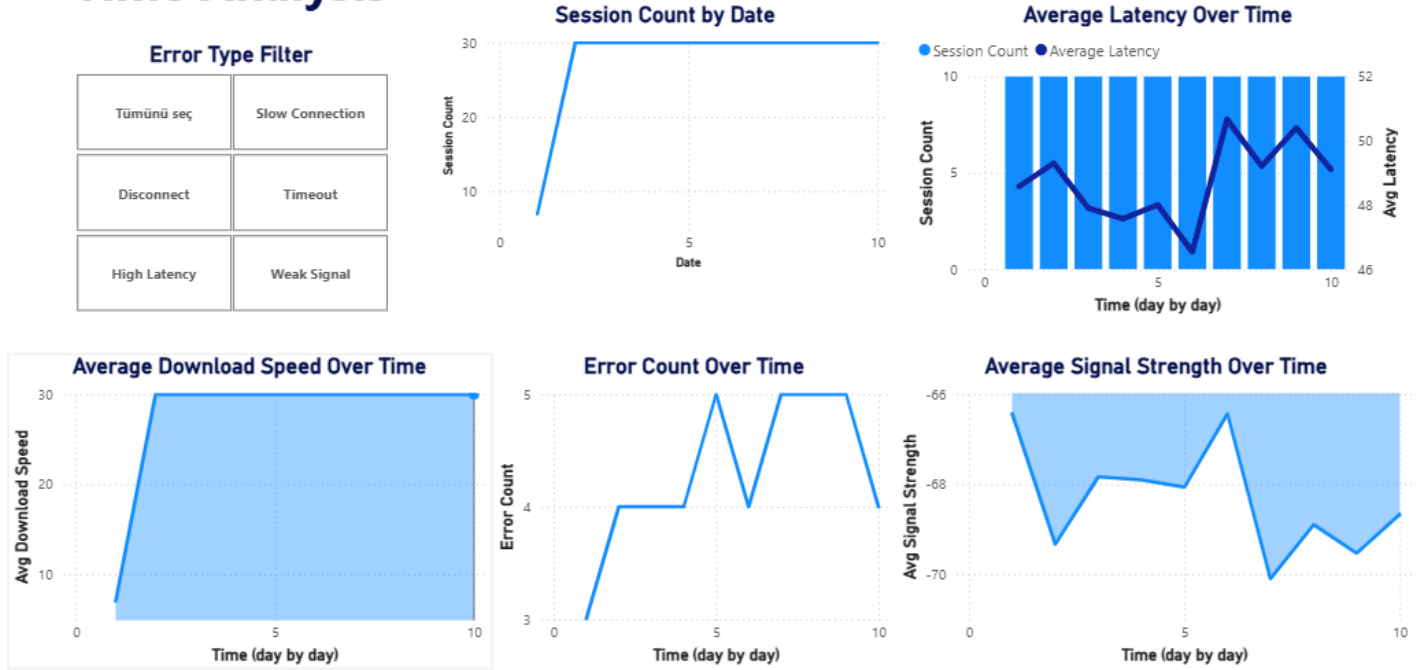
I mainly used the Devices and Users tables. The device_type and brand columns from the Devices table were used to create the device distribution visuals. I also used subscription and user-related information to understand how devices are distributed among users.

The report page includes a pie chart, bar charts, a brand slicer, and a summary table. The pie chart shows the distribution of devices by type, while the bar chart shows the number of devices by brand. The summary table gives a clearer view of device type and brand information. This page helps to understand the device structure of the dataset before analyzing performance and error-related reports.

Report Page 5. Time Analysis

In this report page, I focused on time-based network analysis. The main purpose of this page is to observe how network performance and connection errors change over time.

Time Analysis



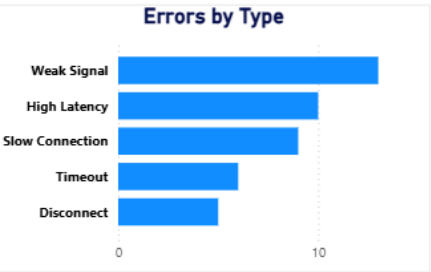
I mainly used the Sessions, Network_Logs, and Errors tables. The start_time, timestamp, and error_time columns were used for time-based visuals. I analyzed session count by date, average download speed over time, average latency over time, average signal strength over time, and error count over time.

The report page includes line charts, an area chart, a combo chart, and an error type slicer. These visuals help to see daily performance changes and possible periods where errors or latency values become higher.

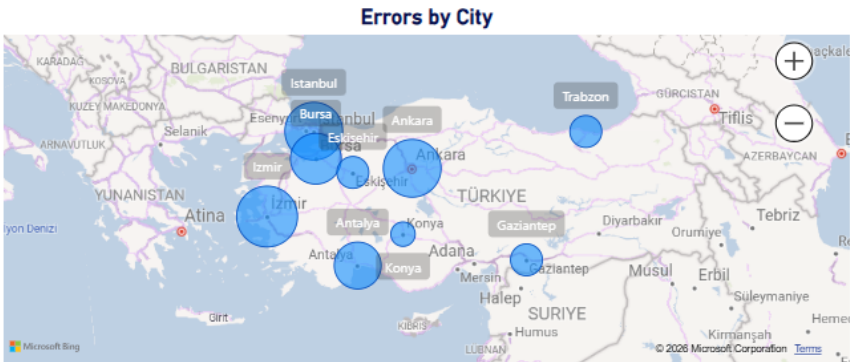
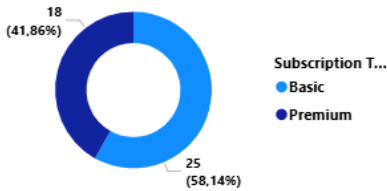
Report Page 6. Error Analysis

In this report page, I focused on connection errors in the dataset. The main purpose of this page is to analyze errors by type, city, subscription type, and router model.

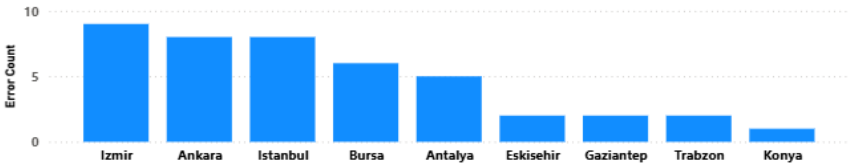
Error Analysis



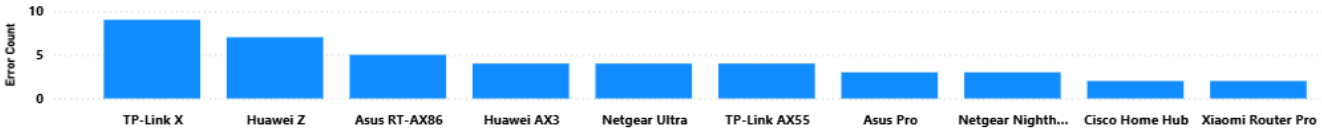
Errors by Subscription Type



Error Count by City



Errors by Router Model



I mainly used the Errors, Sessions, Users, Locations, and Routers tables. The Errors table was used to show the distribution of different error types such as Weak Signal, High Latency, Slow Connection, Timeout, and Disconnect. The Locations table was used to analyze errors by city, while the Users table was used to compare errors by subscription type. Router information was also included to see whether some router models are associated with more error records.

The report page includes bar charts, a map visual, and a donut chart. The map shows the geographical distribution of errors, while the other visuals compare error counts by type, city, router model, and subscription type. This page is useful for understanding where connection problems occur more frequently and which error types are more common.

Report Page 7. Error & Reliability Summary

In this report page, I created a general reliability summary of the dataset. Instead of showing individual session records, this page summarizes the most important performance and error-related values in one table.

Error & Reliability Summary

Error Type Filter		Region Filter		Subscription Type Filter		
Disconnect	Timeout	Aegean	Marmara	Basic		
High Latency	Weak Signal	Black Sea	Mediterranean	Premium		
Slow Connection		Central Anatolia	Southeastern			

City	Session Count	Error Count	Avg Download Speed	Avg Upload Speed	Avg Latency	Avg Signal Strength
Adana	6		57,50	13,67	26,94	-57,83
Ankara	12	8	36,26	8,52	63,19	-75,03
Antalya	12	5	48,79	11,55	47,00	-66,33
Bursa	12	6	43,48	10,33	56,24	-70,88
Eskisehir	7	2	55,19	13,19	37,48	-62,57
Gaziantep	6	2	59,44	14,22	28,67	-59,06
Istanbul	17	8	39,05	9,18	51,09	-71,30
Izmir	14	9	33,46	7,84	67,11	-77,35
Konya	6	1	54,22	13,06	34,39	-62,22
Trabzon	8	2	48,92	11,46	41,50	-65,71
	100	43	45,27	10,72	48,74	-68,48

I used the Locations, Users, Sessions, Network_Logs, and Errors tables. The summary table includes city, session count, error count, average download speed, average upload speed, average latency, and average signal strength. These values help compare cities in terms of both performance and reliability.

I also added slicers for error type, region, and subscription type. These slicers make the summary table interactive and allow the analysis to be filtered from different perspectives. This page works as a final analytical summary because it brings together performance, error, location, and subscription information in a compact form.

Responsibilities in the Project

This project was designed and implemented by me. First, I created the database schema and prepared the tables in MSSQL. Then, I generated and inserted the sample data, wrote the necessary SQL queries, views, and stored procedures. Finally, I connected the database to Power BI and prepared the report pages and dashboards.